

Notes: Rampini and Viswanathan (JF, 2010)

Collateral, Risk Management, and the Distribution of Debt Capacity

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model environment	4
3.1	Agents, technology, and collateral	4
3.2	Limited enforcement and collateral constraints	4
4	Debt capacity, financial slack, and risk management	5
4.1	Debt capacity	5
4.2	Minimum down payment	5
4.3	Loan commitments	5
5	Core theoretical results	5
5.1	Productivity and exhaustion of debt capacity	5
5.2	Downsizing of productive firms	6
5.3	Collateralizability and fragility	6
5.4	Net worth and risk management	6
6	Relation to standard risk-management theory	6
6.1	Froot, Scharfstein, and Stein benchmark	6
6.2	Rampini and Viswanathan's dynamic reversal	6
7	Empirical interpretation	7
8	Figures	7
8.1	Figures 1 and 2: Net worth, financial slack, and collateral multipliers	7
8.2	Figures 3 and 4: Standard hedging theory versus collateral-constrained financing . .	8
9	Short summary	9
10	Conclusion	9
10.1	Core conclusion	9
10.2	Economic intuition	9
10.3	Incremental contribution revisited	10
10.4	Implications for finance and macroeconomics	10

10.5 Limitations 10

10.6 Takeaway 10

1 Big picture

- **Question.** How do collateral constraints jointly shape firm financing, risk management, debt capacity, and investment dynamics? The paper asks why financially constrained firms often hedge less, even though standard risk-management theory predicts that constrained firms should hedge more.
- **Core idea.** Financing and risk management are not separate decisions. Both require promises to pay, and those promises are limited by collateral. Therefore, using collateral capacity to hedge future states reduces the debt capacity available for current investment.
- **Main mechanism.** Conserving debt capacity is costly because it means giving up current investment. This opportunity cost is especially high for firms with low net worth or high productivity because they have high marginal returns to current investment.
- **Central result.** More constrained firms optimally exhaust debt capacity and hedge less. Very constrained firms may not hedge at all. Better-capitalized or less productive firms can afford to keep financial slack and engage in more risk management.
- **Dynamic implication.** In downturns, cash flows are low and investment opportunities may be attractive because capital is cheap. But the most productive or least-capitalized firms may have already exhausted debt capacity, so they cannot seize these opportunities and may have to downsize.
- **Aggregate implication.** Capital can become less productively allocated in downturns. Distributional differences in debt capacity can amplify aggregate productivity shocks.
- **Risk-management implication.** The paper reverses the usual prediction that financially constrained firms should hedge more. Once up-front investment is endogenous and collateral is scarce, investment financing can dominate hedging.
- **Financial innovation implication.** Higher collateralizability raises ex ante borrowing capacity and leverage, but may also reduce ex post net worth and make contractions more severe when bad states arrive.

2 Incremental contribution of the paper

- **From static risk management to dynamic financing-risk management.** The classic Froot, Scharfstein, and Stein (1993) logic takes current investment as given and predicts that constrained firms hedge to protect future investment. Rampini and Viswanathan make current investment endogenous and show that collateral used for hedging competes with collateral used for financing investment.
- **A unified theory of financing and hedging.** The paper shows that debt capacity, financial slack, loan commitments, and derivatives are different ways of allocating scarce pledgeable resources across dates and states.
- **State-contingent collateral constraints.** The model derives state-contingent collateral constraints from limited enforcement. This allows firms to access complete markets subject to collateral limits, so the absence of hedging is a choice, not a missing-market assumption.

- **A resolution of the hedging puzzle.** Empirically, large firms hedge more than small firms, even though small firms are more likely to be financially constrained. The paper explains this by showing that the most constrained firms optimally allocate scarce debt capacity to current investment instead of hedging.
- **Distribution of debt capacity as a macro-finance object.** The paper shifts attention from aggregate debt capacity to its cross-sectional distribution. Who has unused debt capacity in downturns matters for whether capital is reallocated productively.
- **Collateralizability and financial innovation.** The paper highlights a subtle effect: making assets more collateralizable can increase leverage and current investment, but can also make constrained firms more fragile later.
- **Loan commitments as contingent debt capacity.** The paper provides an interpretation of loan commitments as a way to conserve contingent net worth, but shows that arranging such commitments also requires scarce up-front resources.

3 Model environment

3.1 Agents, technology, and collateral

The economy has risk-neutral firms with limited liability and lenders who can commit and provide state-contingent claims. Firms invest in capital and produce future cash flows. They can default and abscond with cash flows plus a fraction of capital, so lenders can seize only a fraction θ of collateral value.

A firm that invests $k(s^t)$ at date t in state s^t receives at date $t + 1$:

$$A(s^{t+1})f_t(k(s^t)) + q(s^{t+1})k(s^t),$$

where $A(s^{t+1})$ is productivity and $q(s^{t+1})$ is the resale price of capital.

Economic message: the productive asset is also the collateral asset. The ability to finance current investment and the ability to hedge future states both depend on how much collateral value can be pledged.

3.2 Limited enforcement and collateral constraints

The key enforcement result is that the dynamic limited-enforcement problem is equivalent to one-period-ahead state-contingent debt subject to collateral constraints:

$$q(s^t)\theta k(s^{t-1}) \geq Rb(s^t),$$

where $Rb(s^t)$ is the promised payment in state s^t .

Interpretation: the firm can promise to pay in each state only up to the collateralizable value of capital in that state. Because the constraint is state-contingent, the firm can in principle hedge, but hedging uses the same collateral capacity as borrowing.

4 Debt capacity, financial slack, and risk management

4.1 Debt capacity

Given capital $k(s^t)$, debt capacity in state s^{t+1} is

$$q(s^{t+1})\theta k(s^t).$$

If the firm promises less than this amount, it keeps financial slack:

$$h(s^{t+1}) = q(s^{t+1})\theta k(s^t) - Rb(s^{t+1}).$$

Thus, $h(s^{t+1}) > 0$ means the firm has conserved debt capacity for that future state.

Interpretation: financial slack is state-contingent unused borrowing capacity. In this model, risk management is the act of preserving net worth or debt capacity for states where the marginal value of funds will be high.

4.2 Minimum down payment

The minimum down payment per unit of capital is

$$p(s^t) = q(s^t) - R^{-1}E[q(s^{t+1}) | s^t]\theta.$$

The required down payment is lower when expected capital appreciation is higher.

Economic message: lending standards are endogenous. If future collateral value is expected to be high, the borrower can pledge more, so the required down payment falls. If future collateral value is expected to decline, borrowing capacity shrinks and deleveraging pressure rises.

4.3 Loan commitments

A loan commitment promises future financing in a specified state in exchange for an up-front fee. In the model, loan commitments are equivalent to conserving contingent debt capacity. They do not avoid collateral constraints; they merely allocate scarce pledgeable capacity across states.

Interpretation: arranging contingent financing is costly because it consumes resources today that could otherwise support current investment.

5 Core theoretical results

5.1 Productivity and exhaustion of debt capacity

With constant returns to scale, the paper defines the return on internal funds when the firm invests with maximal leverage as $R_1(s)$ and $R_2(s)$. A firm invests at time 0 and exhausts debt capacity if

$$E[R_1 R_2] > \max_s \{R R_2(s)\}.$$

Otherwise, the firm stays on the sidelines and conserves net worth for the most productive future state.

Economic message: the more productive the firm is today, the more attractive current investment is, and the more likely the firm is to use all debt capacity immediately rather than save it for the future.

5.2 Downsizing of productive firms

Firms that exhaust debt capacity may be forced to downsize in low-cash-flow states. They could have arranged contingent financing, but doing so would have required sacrificing profitable current investment.

Economic message: constrained firms are not irrationally under-hedged. They rationally choose current scale over future slack. But this choice makes them fragile in downturns.

5.3 Collateralizability and fragility

An increase in collateralizability θ raises ex ante borrowing capacity. However, because firms borrow more up front, they may have lower ex post net worth. Therefore, higher collateralizability can make later contractions more severe for firms that exhaust debt capacity.

Economic message: better pledgeability is not unambiguously stabilizing. It enables more leverage, but can increase the volatility of investment for constrained firms.

5.4 Net worth and risk management

With decreasing returns, the paper shows that firms with sufficiently low net worth do not engage in risk management. Firms with higher net worth hedge progressively more states, and sufficiently well-capitalized firms engage in complete risk management.

Core intuition:

- Low-net-worth firms operate at small scale and have high marginal returns to current investment.
- For these firms, shifting funds to today dominates shifting funds across future states.
- Higher-net-worth firms have lower marginal returns today and can afford to conserve debt capacity for bad or high-marginal-value future states.

6 Relation to standard risk-management theory

6.1 Froot, Scharfstein, and Stein benchmark

The received theory argues that financially constrained firms should hedge because hedging moves funds into states where internal funds are most valuable. In a static setting with current investment fixed, hedging equalizes the marginal value of net worth across future states:

$$\mu_1(H) = \mu_1(L).$$

6.2 Rampini and Viswanathan's dynamic reversal

In this paper, current investment is not fixed. The firm must choose between:

- using collateral capacity to finance current investment; or
- using collateral capacity to preserve future state-contingent net worth.

If the firm is severely constrained, the marginal value of funds today is so high that all collateral constraints bind. Then the firm does not hedge:

$$\mu_1(H) \neq \mu_1(L).$$

Economic message: financial constraints create a desire to hedge, but they also restrict the resources needed to hedge. For the most constrained firms, investment financing wins.

7 Empirical interpretation

Although the paper is theoretical, it is designed to rationalize several empirical patterns.

- **Small firms hedge less.** Small firms are likely to have lower net worth and stronger current financing needs, so they exhaust debt capacity instead of hedging.
- **Large firms hedge more.** Large firms are better capitalized and can afford to allocate some debt capacity to future states.
- **Downturns reallocate capital imperfectly.** Firms with the highest productivity may be the most constrained and unable to buy cheap capital when downturns create investment opportunities.
- **Financial innovation can amplify cycles.** Higher collateralizability can support more leverage in expansions but make constrained firms more vulnerable later.
- **Loan commitments are costly insurance.** Firms with scarce internal funds may not pay up-front fees for future commitments because doing so reduces current investment.

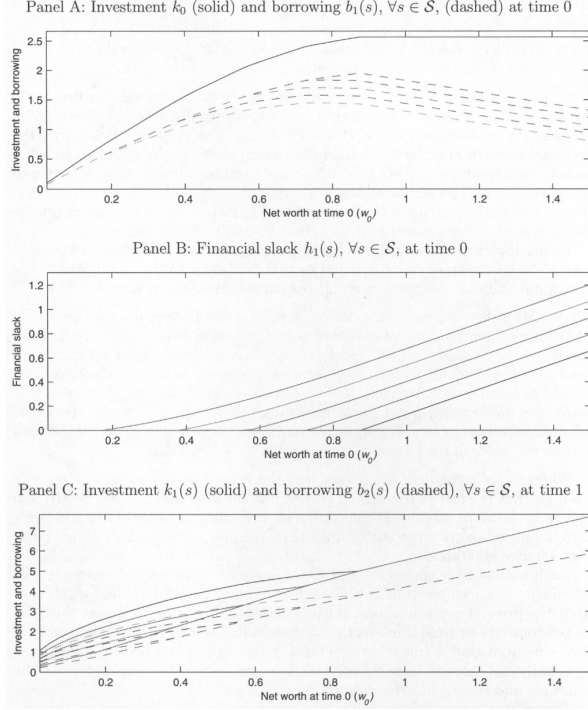
8 Figures

8.1 Figures 1 and 2: Net worth, financial slack, and collateral multipliers

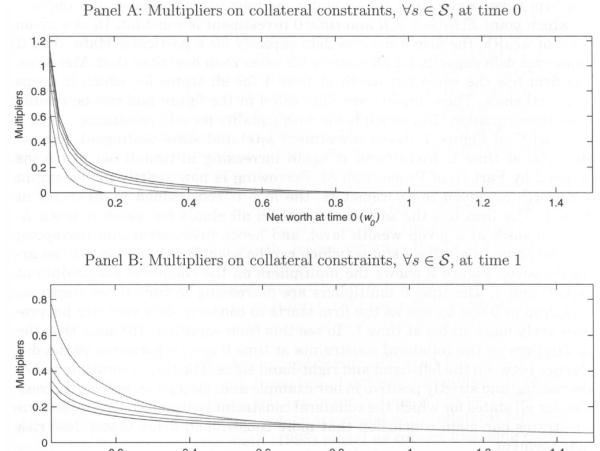
Read:

- Figure 1 shows how investment, state-contingent borrowing, and financial slack vary with initial net worth.
- At low net worth, financial slack is zero: the firm exhausts debt capacity in every future state.
- As net worth rises, the firm starts to keep slack for some states, then for progressively more states.
- Figure 2 shows the corresponding collateral-constraint multipliers. Multipliers are high when net worth is low and fall as collateral constraints become less binding.

Economic message: risk management is increasing in net worth. The most constrained firms use all pledgeable resources for current investment; better-capitalized firms can conserve debt capacity and hedge.



(a) Figure 1: Investment, state-contingent borrowing, and financial slack.



(b) Figure 2: Multipliers on collateral constraints.

8.2 Figures 3 and 4: Standard hedging theory versus collateral-constrained financing

Read:

- Figure 3 illustrates the Froot, Scharfstein, and Stein benchmark: with no current financing need and frictionless complete markets, the firm hedges until the marginal value of net worth is equal across states.
- Figure 4 illustrates the Rampini-Viswanathan mechanism: when there is a current financing need and payments are collateral constrained, financing needs can override hedging.
- In Figure 4, the firm does not equalize marginal values across states because all collateral capacity is absorbed by the need to finance current investment.

Economic message: standard theory asks how a constrained firm should hedge future states; this paper asks whether a constrained firm can afford to hedge after financing current investment. This dynamic distinction changes the prediction.

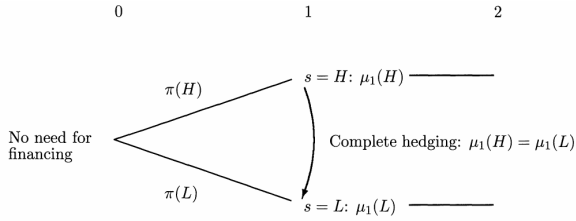


Figure 3. Risk management as in Froot, Scharfstein, and Stein (1993). With complete

(a) Figure 3: Risk management in the Froot-Scharfstein-Stein benchmark.

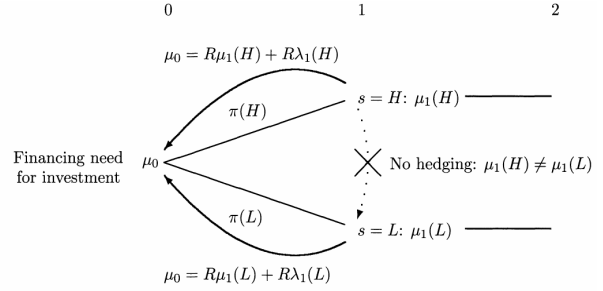


Figure 4. Financing and risk management subject to collateral constraints. With com-

(b) Figure 4: Financing and risk management under collateral constraints.

9 Short summary

- The paper develops a dynamic model of collateralized financing with limited enforcement.
- Collateral constraints limit all promises to pay, including debt, hedges, and loan commitments.
- Financial slack is unused debt capacity; risk management is state-contingent conservation of net worth.
- Conserving debt capacity has an opportunity cost: the firm must give up current investment.
- More productive and lower-net-worth firms face the highest opportunity cost of slack.
- These firms optimally exhaust debt capacity and hedge less, or not at all.
- Better-capitalized firms hedge more because they can afford to allocate resources to future states.
- In downturns, productive constrained firms may be unable to invest in cheap capital and may downsize.
- Higher collateralizability can raise leverage and increase ex post fragility.
- The paper resolves the empirical puzzle that small constrained firms hedge less than large firms.

10 Conclusion

10.1 Core conclusion

Rampini and Viswanathan show that financing and risk management are fundamentally linked through collateral. Because both borrowing and hedging require credible promises to pay, and credible promises are limited by collateral, firms cannot choose financing and risk management independently.

10.2 Economic intuition

A constrained firm faces a scarce-resource allocation problem. It can use collateral capacity today to finance investment, or it can preserve some capacity to protect future states. For firms with high

marginal returns to current investment, the current investment margin dominates. These firms exhaust debt capacity and leave themselves exposed to future shocks.

10.3 Incremental contribution revisited

The paper's main contribution is to turn corporate risk management into a dynamic collateral-allocation problem. Standard risk-management theory says that constrained firms should hedge because future internal funds are valuable. This paper says that the same constraints that make future funds valuable also make hedging costly. Therefore, the most constrained firms may hedge the least.

10.4 Implications for finance and macroeconomics

The distribution of unused debt capacity matters. In a downturn, the firms best able to use investment opportunities may be precisely the firms that lack slack. This can reduce allocative efficiency and amplify aggregate shocks. The model also suggests that financial innovation and greater pledgeability may increase both borrowing capacity and vulnerability.

10.5 Limitations

The paper is a theoretical model and abstracts from many institutional features of real corporate finance, including taxes, agency conflicts beyond limited enforcement, fixed hedging costs, and endogenous asset prices. The model also does not estimate the size of collateral constraints empirically.

10.6 Takeaway

The central takeaway is that not hedging can be an optimal consequence of being severely financially constrained. For the most constrained firms, financing current investment is more urgent than preserving debt capacity for future states. This insight links capital structure, risk management, and macroeconomic vulnerability in one framework.